

Project Name: *MovieIQ Data Analysis*

Null Hypothesis (H_0)

There is **no significant difference** in popularity between successful and unsuccessful movies.

Alternative Hypothesis (H_1)

There **is** a significant difference in popularity between successful and unsuccessful movies.

Decision Rule

- If **p-value** < **0.05** → Reject H_0 .
- If **p-value** $\geq$ **0.05** → Fail to reject H_0 .

Chi-Square Test

Null Hypothesis (H_0)

Movie genres and movie success are independent.

Alternative Hypothesis (H_1)

Movie genre and movie success are associated.

Explain the p-value

A p-value measures how likely it is to observe the obtained results if the null hypothesis is true. A small p-value (less than 0.05) indicates strong evidence against the null hypothesis, so we reject it. A larger p-value suggests there is not enough evidence to reject the null hypothesis.

